SLVMEval: Synthetic Meta Evaluation Benchmark for Text-to-Long Video Generation

Anonymous CVPR submission

Paper ID 9353

Abstract

We introduce SLVMEval, a benchmark for meta-evaluating text-to-video (T2V) evaluation systems. SLVMEval focuses on assessing these systems on long videos of up to 10,486 seconds (approximately 3 hours). Our benchmark targets a fundamental requirement: whether systems can accurately judge video quality in settings that are easy for humans to assess. We adopt a pairwise comparison-based meta-evaluation framework. Building on dense video captioning datasets, we synthetically degrade source videos to create controlled "high-quality vs. low-quality" pairs across 10 distinct aspects. We then use crowdsourcing to filter and retain only those pairs in which the degradation is clearly perceptible, thereby establishing the final testbed. Using this testbed, we assess the reliability of existing evaluation systems in ranking these pairs. Our experiments show that human evaluators identify the better long video with 84.7%–96.8% accuracy, while in 9 of the 10 aspects, the accuracy of these systems falls short of human judgment, revealing weaknesses in text-to-long video evaluation.

1. Introduction

Much of the video content we consume daily, especially movies and TV series, consists of long-form content that typically spans tens of minutes to several hours. Despite recent advances in text-to-video (T2V) models [4, 9], their outputs remain limited to a few minutes, and generating long videos at movie scale continues to pose technical challenges. In response, systems that can, in principle, generate arbitrarily long videos have recently emerged [5, 38], highlighting long-form video generation as the next frontier.

Despite the momentum in text-to-long video (T2LV) research, validation of evaluation metrics for T2LV has fallen behind. Currently, researchers often use metrics such as VideoScore [11] to evaluate T2LV results. However, such metrics were originally designed for short videos lasting from a few to tens of seconds, and therefore, T2LV results may not be assessed appropriately due to a mismatch in video length. Moreover, even if we aim to validate whether such a mismatch exists using human-annotated datasets, such as VBench [15] and UVE [26], these datasets cannot reliably assess evaluation metrics for T2LV, as they also focus exclusively on short videos. In this context, developing a meta-evaluation benchmark for T2LV is a crucial step toward advancing T2LV research.

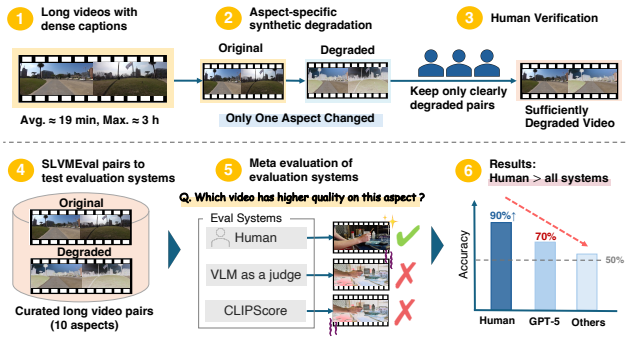

Figure 1. Overview of SLVMEval. We construct human-validated pairs of original and aspect-specifically degraded long videos, and test various automatic evaluation systems. Although human evaluators reliably pick the better video, all current automatic evaluation systems still lag behind human performance on most aspects, revealing critical weaknesses in T2LV evaluation.

To this end, we propose the Synthetic Long-Video Meta-Evaluation Benchmark (*SLVMEval*), a new benchmark for meta-evaluating automatic evaluation systems for T2LV. The purpose of our benchmark is to meta-evaluate how well evaluation systems meet the minimum requirements for assessing long videos, specifically whether they can make judgments that humans can easily make. Figure 1 illustrates an overview. SLVMEval comprises pairs of high-quality and low-quality videos. High-quality videos are sampled from existing dense video captioning datasets, and low-quality videos are synthesized by degrading them with respect to specific aspects and then manually filtered. The resulting videos range from several minutes to several hours,

which is well beyond the durations generated by current T2V models. We use SLVMEval to evaluate several baseline automatic evaluation systems, such as VLM-as-a-judge and CLIPScore. In human evaluation, crowd workers distinguish the high-quality video from the low-quality one with 84.7%–96.8% accuracy. In contrast, existing evaluation systems lag human evaluators on nine of the ten aspects. This gap is especially large for evaluation aspects that require semantic and temporal consistency between the video and the prompt. Moreover, for almost all automatic evaluation systems, we observe a decrease in accuracy as video duration increases, suggesting that existing evaluation systems have some limitations when evaluating long videos.

Furthermore, when we compare evaluation results on our benchmark obtained with and without applying crowd-worker-based filtering to the test data, we observe a strong correlation between them. This suggests that our methodology can be used to construct and extend the benchmark without always relying on costly manual data filtering. [1]

2. Related work

Text-to-video generation. Research on T2V models has recently gained significant attention. They can be broadly categorized into two main approaches: T2V models based on diffusion models [10, 17] and those based on auto-regressive transformers [7]. Among diffusion-based T2V models, Sora [4] supports video generation of up to 60 seconds, and Kling [20] supports up to 2 minutes. Furthermore, StreamingT2V [12] enables the generation of highly consistent videos exceeding two minutes by constructing a pipeline that generates videos auto-regressively in chunk-based units. Phenaki [38] is an auto-regressive transformer-based T2V model that takes discretized video tokens as input and outputs them autoregressively. By performing the next video token prediction, it is theoretically capable of generating videos of arbitrary lengths. However, there is a limitation that only 1.4-second videos are used to train this model. This limitation highlights that generating coherent, long videos remains a significant challenge. Our framework aims to construct an environment that evaluates automatic evaluation systems in advance, which will be necessary for developing T2LV models in the future.

Meta-evaluation of T2V evaluation systems. Numerous benchmarks have been proposed for evaluating T2V models [15, 24–26]. VBench [15] provides fine-grained human annotations along dimensions such as object fidelity, aesthetics, and technical quality; its successors, VBench++ [16] and VBench2 [45], extend this coverage to additional aspects and model families. UVE-Bench [26], in contrast, centers on language-model-based evaluation, offering human pairwise-comparison data to gauge the alignment between automatic judges and human preference. Despite their merits, these resources are confined to clips of roughly 10 seconds. To transcend this limitation, we construct a new meta-evaluation testbed for long videos, i.e., up to several hours, thereby enabling the development of automatic metrics suited to future T2V systems.

Long-video understanding with VLMs. One of the goals of VLMs is to understand long video content that humans watch daily, and their development for processing such videos is advancing [40]. For example, InternVideo2.5 [6] can process videos of several minutes to hours. The long-video understanding capabilities of these VLMs are predominantly evaluated through Visual Question Answering (VQA) tasks [28, 34, 40, 42, 46]. MLVU [46] is a benchmark that evaluates content understanding abilities on long videos, including movies, surveillance footage, and first-person view videos, through diverse tasks such as topic inference, anomaly detection, and summarization. This benchmark has demonstrated that VLM performance on long videos deteriorates significantly as video duration increases; for instance, Video-LLaMA-2 [44] performance dropped to chance level when videos were extended from 3 to 10 minutes. However, these long-video benchmarks focus on content understanding rather than meta-evaluating automatic evaluation systems for text-to-video generation, and they do not assess whether such systems can detect controlled quality degradations that are easily recognizable to humans. In contrast, we introduce a synthetic long-video meta-evaluation benchmark that uses controlled "high-quality vs. low-quality" pairs to test whether automatic evaluation systems for T2V models can reliably judge both *Video Quality* and *Video-Text Consistency* on long videos.

3. Meta-evaluation framework

We adopt a pairwise comparison approach for our meta-evaluation framework, following the practice in VBench and UVE [15, 26].

3.1. Evaluation system

Let $\mathcal{P}$ denote the set of textual prompts and $\mathcal{V}$ the set of all possible videos. We regard a T2V model as a mapping $g : \mathcal{P} \to \mathcal{V}$. For any prompt $p \in \mathcal{P}$, an evaluation system e receives the pair of videos $\{u_p, v_p\} = \{g(p), g'(p)\}$ produced by two T2V models g and g', and returns the one it judges to be of higher quality, i.e., $e(p, \{u_p, v_p\}) = z$, where $z \in \{u_p, v_p\}$.

[1] All code and data will be publicly released upon acceptance.

3.2. Meta-evaluation criteria

As a metric for pairwise comparison, we measure an evaluation system's reliability with accuracy, the proportion of video pairs for which the system selects the better video. To compute this metric, we build a test set $\mathcal{D} = \{(p_i, \{v^+_{p_i}, v^-_{p_i}\})\}_{i=1}^N$, where each prompt p_i has two generated videos: $v^+_{p_i}$ (higher quality) and $v^-_{p_i}$ (lower quality).

$$\mathrm{acc}(e, \mathcal{D}) = \frac{1}{|\mathcal{D}|} \sum_{(p,\{v^+_p, v^-_p\})\in\mathcal{D}} \mathbf{1}\big[e(p, \{v^+_p, v^-_p\}) = v^+_p\big], \quad (1)$$

where $\mathbf{1}[\cdot]$ is the indicator function that returns 1 when the evaluation system selects v^+_p and 0 otherwise. A higher accuracy indicates that the evaluation system is better at judging video quality from a particular aspect.

4. Dataset construction

Our goal is to meta-evaluate whether existing evaluation systems meet the minimum requirements to correctly assess the quality of generated videos. To this end, we construct synthetic video pairs that differ only in one specified aspect of quality and use them to build a test set for fine-grained pairwise evaluation.

4.1. Dataset definition

We construct a synthetic test set $\mathcal{D}$ that probes whether an evaluation system can correctly judge video quality along several fundamental aspects. Starting from a manually curated prompt-video corpus $\mathcal{D}_{\text{src}} = \{(p_i, v^+_{p_i})\}_{i=1}^N$, we generate, for each aspect a, a degraded counterpart for every original video and construct the corresponding aspect-specific dataset, $\mathcal{D}_a$, namely:

$$\begin{aligned} &\mathcal{D}_a = \{(p, \{v^+_p, v^-_p\}) \mid (p, v^+_p) \in \mathcal{D}_{\text{src}}\}, \\ &\text{where} \quad v^-_p = \Phi^{\text{low}}_a(p, v^+_p). \end{aligned} \quad (2)$$

The function Φ^{low}_a takes a pair (p, v^+_p) and returns a degraded video $v^-_p \in \mathcal{V}$ for aspect a. For each aspect we examine, we apply a different function and build a separate test set for that aspect. Details of the degrading operation are provided in Section 4.3.

4.2. Dataset source

We select the Vript dataset [43] as our dataset source. We choose Vript because it comprises human-made long videos that align with the focus of our study. Furthermore, each video in Vript is segmented into semantically coherent clips and annotated with descriptive captions since it is designed for the dense video captioning task [19]. For evaluation, we sample each video at 1 fps and resize every frame so that its longer side measures 512 pixels.

Let $v \in \mathcal{V}$ denote a video with T frames, and let $f_{v,t}$ be its t-th frame. If the video splits into M clips, we represent the m-th clip as $c_{v,m} = (f_{v,t})_{t\in[t_m, t_{m+1})}$, where t_m and t_{m+1} mark the first frames of the m-th and $(m+1)$-th clips, respectively. We assign a caption p_m to each clip $c_{v,m}$ and define the prompt for v as the concatenation $p = p_1 \oplus p_2 \oplus \cdots \oplus p_M$.

Algorithm 1 Overview of Φ^{low}_a

Require: Prompt $p \in \mathcal{P}$, Video $v_p \in \mathcal{V}$
Ensure: v^-_p

```
S ← SampleIdx(M_{v_p}, 5)   ▷ random sample 5 clips
F ← []
for m ← 0 to M_{v_p} − 1 do
    clip ← c_{v_p,m}
    if m ∈ S then
        F ← F ∪ DegradeClip(clip, p_m)
    else
        F ← F ∪ clip
    end if
end for
v_p^- ← Concat(F)
return v_p^-
```

4.3. Degrading operations

Following prior work, we define multiple evaluation aspects. However, because systems for generating long videos are still in their early stages, it is difficult to identify these aspects solely from the failure modes observed in such videos. Therefore, using benchmarks for short videos as references, we define 10 evaluation aspects grouped into two main categories: **Video Quality** and **Video-Text Consistency**. Using these aspects, we meta-evaluate whether an evaluation system possesses at least the minimal capabilities required for T2LV evaluation.

This categorization is inspired by comprehensive benchmarks like VBench [15], but we have also adapted aspects of long-range temporal alignment, such as Temporal Flow and Comprehensiveness, to probe the specific challenges of T2LV generation. For each aspect, we independently invoke DegradeClip to create a degraded video v^-_p (pseudocode appears in Algorithm 1). Figure 2 shows example degraded outputs. In the following, we introduce every aspect and then explain the associated degrading operation.[2]

4.3.1. Video Quality

This category includes four aspects that evaluate the visual appearance.

Aesthetics. We assess whether the evaluation system can judge frame-level aesthetic quality [15]. To create the degraded video v^-_p, we reduce the contrast of selected clips in the original video.

[2] See the supplementary material for further implementation details.

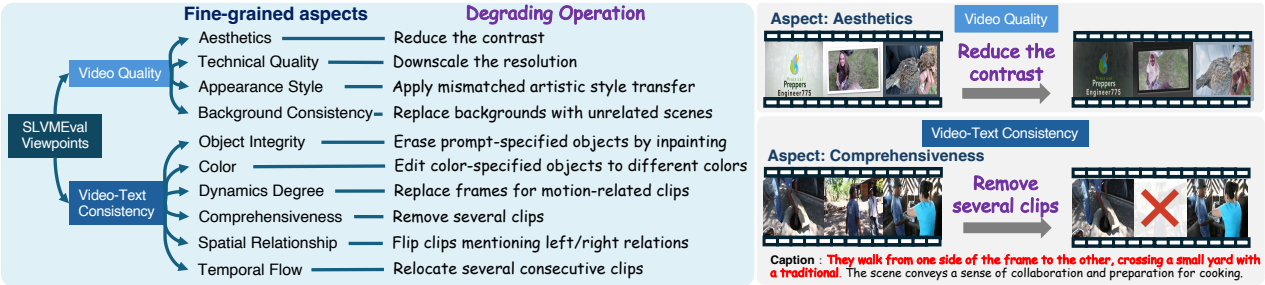

Figure 2. SLVMEval viewpoints and aspect-specific degrading operations. We organize the benchmark into two groups, *Video Quality* and *Video-Text Consistency*, and define 10 aspects. For each aspect, we construct paired videos by applying a controlled synthetic degradation to the original long video while keeping all other factors unchanged. The lower panels show example pairs. These controlled pairs enable precise meta-evaluation of whether an automatic evaluator can reliably identify the high-quality video under each viewpoint. We provide additional example pairs in the supplementary material.

Technical Quality. We assess whether the evaluation system can detect low-resolution artifacts [26]. To create v_p^-, we downscale the resolution of selected clips from the original video.

Appearance Style. We assess whether the evaluation system can detect unnatural video style. To create v_p^-, we convert all frames in selected clips to a different style (e.g., oil painting, manga, watercolor, sketch) using the style-transfer API in OpenCV [3].

Background Consistency. We assess whether the evaluation system can determine if the background remains visually stable across frames. To create v_p^-, we remove the original backgrounds with rembg [8] and replace them with randomly sampled landscape images from the nature-dataset [27].

4.3.2. Video-Text Consistency

This category covers aspects that rely on video semantics. We adopt six aspects that test whether the system can assess consistency between the video and its corresponding text prompt.

Temporal Flow. We assess whether the evaluation system can follow the order of events in a video. Specifically, we create v_p^- by moving five consecutive clips to random positions, thereby breaking the original event order.

Comprehensiveness. We assess whether the evaluation system can verify that a video fully covers the prompt. To create v_p^-, we randomly remove five clips from the original video.

Object Integrity. We assess whether the evaluation system can recognize the presence of prompt-specified objects. To create v_p^-, we first extract object names from each prompt p_m with Qwen3-8B [35], then localize them using Grounding DINO [23], and finally erase them via Stable-Diffusion-Inpainting [30].

Spatial Relationship. We assess whether the evaluation system can determine if the spatial relations specified in the prompt (left/right) match each frame. To create v_p^-, we first use Qwen3-8B to identify clips whose captions mention left/right relations of some object, and then horizontally flip all frames in those clips.

Dynamics Degree. We assess whether the evaluation system can determine if the amount of motion specified in the target prompt is appropriate. To create v_p^-, we first use Qwen3-8B to identify clips whose captions mention object motion, and then replace every frame in those motion-involving clips with the middle frame of the clip, producing a static clip with no motion.

Color. We assess whether the evaluation system can judge if the colors specified in the prompt match the contents of the generated video. To create v_p^-, we use Qwen-Image-Edit-2509 [41] to modify the colors of objects in all frames of the relevant clips. We use Qwen3-8B to identify which clips mention an object's color.

4.4. Human-Annotated Dataset Filtering

To ensure the quality and validity of the synthesized degraded video pairs, we conduct human annotations and filter the dataset accordingly. The goal is to verify that each degradation is applied as intended and is clearly perceptible to humans.

Annotation Rules. We show crowd workers the target aspect, the paired videos $\{v_p^+, v_p^-\}$, and a description of the applied degradation. For each pair, five crowd workers rate the success of the degradation on a three-point A/B/C scale.

A. The degradation clearly succeeds in all selected clips.
B. The degradation succeeds in at least one but not all selected clips, or the effect is weak.
C. The degradation fails in all selected clips.

Dataset filtering. Based on the ratings of the five crowd workers, we select samples to include in the final test set (i.e., those judged to have undergone successful degradation). We use the following success conditions:

Table 1. Statistics of existing and our benchmarks. VBench-Long [16] provides only a framework for automatic evaluation and does not include human annotations of video quality. It supplies prompts as inputs to T2V models, assuming generated videos of approximately one minute in length [14].

	Human Annotation	Aspects	Videos	Unique Prompts	Max Duration (sec)	Avg. Prompt Length (chars)
UVE-Bench [26]	✓	15	1,045	293	6.1	73.68
VBench [15]	✓	16	21,110	968	3.3	41.32
VBench Long [16]		16	N/A	944	N/A	41.00
SLVMEval (**ours**)	✓	10	3,932	1,461	10,486.0	57,883.52

1. **No C votes.**: We exclude samples for which any crowd worker judged the degradation a complete failure.
2. **Votes for A exceed votes for B.**: We retain only samples for which the crowd worker agreement leans toward "clear success" rather than "partial success".

This filtering ensures that each sample exhibits a degradation that humans clearly agree on.[3]

4.5. Dataset statistics

For each evaluation aspect, we binned all candidate videos by duration in 100-second increments and randomly sampled up to 20 videos from each bin. We then performed manual filtering and retained only videos in which the degradation worked as intended. In total, 3,932 videos remained. The resulting clips average 1141 seconds ($\simeq$ 19 minutes) in length, with the longest reaching 10,486 seconds ($\simeq$ 2 hours and 54 minutes). We also confirm that the dataset spans 15 distinct content categories in a balanced manner. [4]

Table 1 compares our dataset with existing meta-evaluation datasets. Compared with manually annotated resources such as VBench [15] and UVE [26], our collection features significantly longer prompts and substantially longer videos, providing a more challenging testbed for long-video evaluation.

5. Baseline automatic evaluation systems

5.1. Pairwise VLM-as-a-judge

Recently, several studies have proposed using VLMs as evaluation systems [21, 22]. To assess their effectiveness for T2LV evaluation, we implement two variants of VLM-as-a-judge: **video-based evaluation** and **text-based evaluation**. For both systems, we use GPT-5 and GPT-5-mini [29] as proprietary models, and Qwen3-VL-235B-A22B-Thinking [1, 2, 35, 39] as an open-weight model.

Video-based evaluation. Given a prompt p and two candidate videos $u_p, v_p \in \mathcal{V}$, the VLM predicts which video is better with respect to aspect a. Formally,

$$e_a(p,\{u_p,v_p\})=\begin{cases} u_p, \text{ if } \mathrm{VLM}(a,p,u_p,v_p)=\text{“first”}, \\ v_p, \text{ if } \mathrm{VLM}(a,p,u_p,v_p)=\text{“second”}. \end{cases} \quad (3)$$

Here, VLM takes as input the prompt p, the videos u_p and v_p, and the evaluation aspect a, and outputs either "first" or "second" to indicate which video is preferred.

Text-based evaluation. In this setting, the VLM first generates captions $caption_{u_p}$ and $caption_{v_p}$ for u_p and v_p, respectively. Then a language model (LM) compares these captions with the prompt p and selects the caption that aligns more closely with the prompt; the corresponding video is returned as the final output. Formally,

$$\begin{aligned} d &= \mathrm{LM}(a, p, \mathrm{VLM}_{\mathrm{cap}}(a, u_p), \mathrm{VLM}_{\mathrm{cap}}(a, v_p)), \\ e_a(p, \{u_p, v_p\}) &= \begin{cases} u_p, & \text{if } d = \text{“first”}, \\ v_p, & \text{if } d = \text{“second”}. \end{cases} \end{aligned} \quad (4)$$

Here, $\mathrm{VLM}_{\mathrm{cap}}$ is a video-captioning model that generates a caption for the input video with respect to aspect a. LM takes the prompt p and the generated captions $caption_{u_p}$ and $caption_{v_p}$ as input, and returns a decision $d \in \{\text{“first”}, \text{“second”}\}$ indicating which caption better matches the prompt. For $\mathrm{VLM}_{\mathrm{cap}}$, we use GPT-5-mini in the closed setting and Qwen3-VL-235B-A22B-Thinking in the open-weight setting. This approach can be viewed as a form of round-trip evaluation [32], and similar metrics are used in BLIP-BLEU and in the EvalCrafter baseline system [25]. We expect that projecting a video into text space can make aspects that are hard to judge from raw frame sequences easier to evaluate.

For the GPT-5 family, context-length limits prevent us from feeding all frames to the model. We thus automatically detect clip boundaries with FFmpeg [37] and extract the center frame from each clip. We feed these frames to the VLM as a concise summary of the video.[5] To mitigate LM-specific biases, such as position bias [36], we also evaluate each pair twice by swapping the order of u_p and v_p.[6]

[3] The supplementary material details the annotation platform (Yahoo! Crowdsourcing), compensation, and quality control.

[4] Detailed distributions for both video duration and content categories are provided in the supplementary material.

[5] The source dataset also includes clip-boundary annotations. We do not

Table 2. Accuracy (%) of each baseline system on SLVMEval. Numbers are accuracy % ± 95% CI (percentage points). Blue bold = best per aspect; green = second best. The chance level is 50%.

	Video Quality				Video-Text Consistency					
	Aesthetics	Technical Quality	Appearance Style	Background Consistency	Object Integrity	Color	Dynamics Degree	Comprehen-siveness	Spatial Relationship	Temporal Flow
Video-based										
GPT-5	90.1±2.5	85.8±4.2	88.9±2.5	**98.9**±0.8	72.0±6.2	84.3±3.5	35.3±3.6	51.3±4.5	59.7±4.4	50.3±4.1
GPT-5-mini	84.0±3.0	48.1±6.1	78.0±3.2	95.2±1.6	66.5±6.5	69.4±4.5	31.5±3.5	45.7±4.5	51.1±4.5	43.7±4.1
Qwen3	55.7±4.1	51.9±6.1	55.3±3.9	49.7±3.7	38.5±6.7	48.8±4.9	50.0±3.8	52.6±4.5	51.7±4.5	50.2±4.1
Text-based										
GPT-5	74.8±3.6	46.2±6.1	81.1±3.1	83.8±2.7	68.0±6.5	68.9±4.5	43.1±3.8	50.6±4.5	47.0±4.5	43.5±4.1
GPT-5-mini	75.0±3.6	53.8±6.1	79.6±3.2	81.1±2.9	65.5±6.6	71.8±4.4	43.8±3.8	50.6±4.5	51.1±4.5	41.2±4.0
Qwen3	51.6±4.1	50.0±6.1	72.4±3.5	73.0±3.3	51.0±6.9	61.0±4.7	52.7±3.8	51.7±4.5	50.2±4.5	55.6±4.1
CLIPScore	56.4±5.8	72.3±7.7	53.2±5.5	68.6±4.8	76.0±8.4	66.2±6.5	51.7±5.4	57.4±6.3	55.1±6.3	50.5±5.8
VideoScore	52.5±5.8	33.8±8.1	65.7±5.3	71.2±4.7	66.0±9.3	33.8±6.5	48.6±5.4	34.5±6.1	49.6±6.4	46.3±5.8
Human	**96.5**±2.1	**91.8**±4.7	**95.2**±2.4	95.0±2.3	**86.6**±6.7	**96.8**±2.4	**95.9**±2.1	**84.7**±4.6	**88.2**±4.1	**86.6**±4.0

5.2. CLIPScore

As one of the existing evaluation systems, we use the CLIPScore [13]. CLIPScore is an automatic evaluation system based on the similarity between text and visual content, widely used in T2V tasks [31, 33].

In our framework, the CLIPScore-based system is defined as follows:

$$e\big(p, \{u_p, v_p\}\big) = \underset{s \in \{u_p, v_p\}}{\arg\max} \frac{1}{N'_s} \sum_{i=1}^{N'_s} \mathrm{CLIPScore}\left(f^{\mathrm{mid}}_{s,i}, p\right)$$
$$\text{where} \quad f^{\mathrm{mid}}_{s,i} = f_{s, \lfloor \frac{t_i + t_{i+1}}{2} \rfloor}. \tag{5}$$

Here, N'_s is the number of clips detected in video s. We use FFmpeg to segment the videos as described in Section 5.1. $\mathrm{CLIPScore}\left(f^{\mathrm{mid}}_{s,i}, p\right)$ is the CLIPScore between the center frame of the i-th detected clip and the prompt p. We use Jina CLIP v2 [18] as the base model; it can handle text inputs of up to 8192 tokens.

5.3. VideoScore

We use VideoScore-v1.1 [11] as one of our baseline evaluation systems. VideoScore is a VLM-based automatic evaluation system. It extends a pre-trained VLM by adding a regression head, and it is further fine-tuned on the VideoFeedback dataset [11], which contains human ratings of video quality. VideoScore can assign a quality score to a video from five aspects. Since the aspects we define differ from those defined by VideoScore, we establish a mapping between our aspects and those of VideoScore.[7] In our framework, we define the VideoScore-based system as

$$e_a(p, \{u_p, v_p\}) = \underset{s \in \{u_p, v_p\}}{\arg\max} \ \mathrm{VideoScore}_a(s, p). \tag{6}$$

Here, $\mathrm{VideoScore}_a(s, p)$ denotes the VideoScore for video s and prompt p under aspect a. Since VideoScore can handle at most 48 frames, as in experiments with the GPT-5 family, we provide the central frame of each scene split by FFmpeg as a representative frame to VideoScore. If the sequence length is still exceeded, we randomly sample up to 48 frames from each video.

6. Experiments

Our experiment aims to evaluate the robustness of current evaluation systems when they are applied to long videos.

6.1. Experimental setup

We evaluated the 8 evaluation systems described in Section 5: Pairwise VLM-as-a-judge (video-based, text-based), CLIPScore, and VideoScore. As the meta-evaluation metric, we used accuracy, defined in Section 3.2, which is calculated as the proportion of video pairs where a system correctly identifies the original video as having higher quality.

As a comparison to automatic evaluation systems, we also measured the performance of humans tasked with judging video quality. Using the same interface as for the evaluation systems, we presented each worker with a pair of videos and the corresponding prompt, asked which video they considered to be of higher quality, and computed their accuracy in the same manner.[8]

6.2. Results and discussion

In this section, we discuss the observations and insights we draw from our comprehensive evaluation experiments.

Benchmark validity. As shown in Table 2, human evaluators achieve 84.7%–96.8% accuracy across all 10 aspects.

use these annotations because ground-truth boundaries are not available in a realistic T2V model evaluation scenario.

[6] The evaluation prompts are listed in the supplementary material.

[7] Please refer to the supplementary material for the mapping rules.

[8] The experimental setup details (e.g., the computational resources, human evaluation processes) are described in the supplementary material.

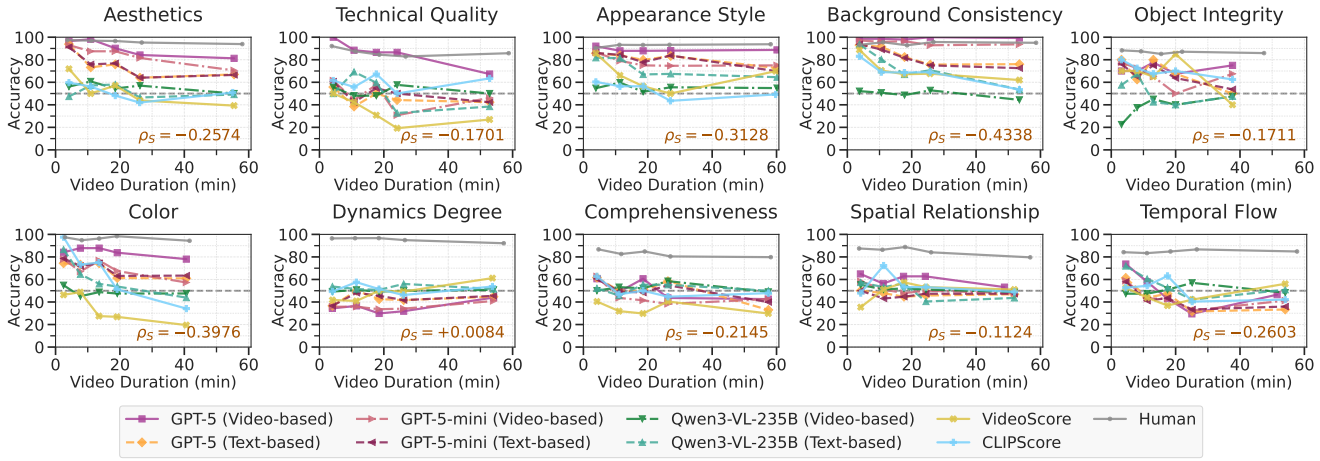

Figure 3. Relationship between video duration and accuracy. We first sort the dataset by video duration, divide it into 4 bins (intervals), and compute accuracy within each bin. The x-axis represents the average video duration in each bin, and the y-axis represents the corresponding accuracy. In addition, for each aspect and each automatic evaluation system (excluding Human), we compute the Spearman rank correlation coefficient ρ_S between video duration and accuracy. Concretely, we sort samples by video duration, divide them into 50 bins, compute accuracy within each bin, and then measure Spearman's ρ_S using these 50 accuracy values. Per-aspect Spearman' s ρ_S values and associated p-values for each automatic evaluation system are reported in the supplementary material.

This indicates that the differences between "high-quality" and "low-quality" videos, as defined by SLVMEval, are easily perceptible to humans. Therefore, our benchmark has an appropriate difficulty for testing our core question: whether an evaluation system satisfies the basic capabilities required to assess video quality.

Overall trends. As shown in Table 2, existing evaluation systems lag human evaluators by between 6.3% and 43.2% on 9 of the 10 aspects. Although these tasks are easy for humans, current systems still struggle to assess long videos on many aspects. Within the Video Quality category, GPT-5 (Video-based evaluation) exceeds 85% accuracy on all aspects, indicating a strong ability to detect surface-level degradations (for example, frame-level visual issues or stylistic shifts), yet it still falls short of human performance. By contrast, in the Video-Text Consistency category, which requires semantic alignment and temporal consistency between the prompt and the video content, accuracies are lower than in the Video Quality category. These results suggest that even the strong proprietary models have difficulty with semantic alignment and temporal consistency in long videos. For text-based evaluation, Qwen3 obtained higher scores than in video-based evaluation, with gains of 23.3 points in Background Consistency, 17.1 points in Appearance Style, and 12.5 points in Object Integrity. These results suggest that text-based evaluation may serve as a useful option for improving evaluation performance for certain models and aspects. CLIPScore ranks second only to human judgments on Object Integrity and Comprehensiveness. We attribute this to CLIP's contrastive pretraining, which yields strong text–image alignment and makes it especially sensitive when objects or clips explicitly mentioned in the prompt are missing. By contrast, because CLIPScore processes frames independently, it remains near chance level on aspects that require reasoning over adjacent frames, such as Dynamics Degree and Temporal Flow, which follows from the system's design. VideoScore falls below the chance level on some aspects. We suspect this is because its five predefined criteria do not align one-to-one with our aspects, leading to inconsistent judgments.

Sensitivity to video duration. Figure 3 shows, for each aspect, how each system's accuracy on our benchmark varies with video duration. We partition the dataset into subsets based on video duration and compute the accuracy within each subset. Then, we compute the Spearman rank correlation coefficient ρ_S between video duration and accuracy. We observe that, for all aspects except Dynamics Degree, the accuracy of most evaluation systems tends to decrease as video duration increases. This pattern is especially clear for aspects such as Background Consistency and Color. In contrast, for aspects on which all systems perform poorly (e.g., Dynamics Degree), the correlation between accuracy and video duration is relatively weak, because the systems often fail even on short videos. Consequently, the p-values, averaged over aspects and systems, do not indicate statistical significance at the 0.05 level, and the results should be interpreted as describing only a general tendency

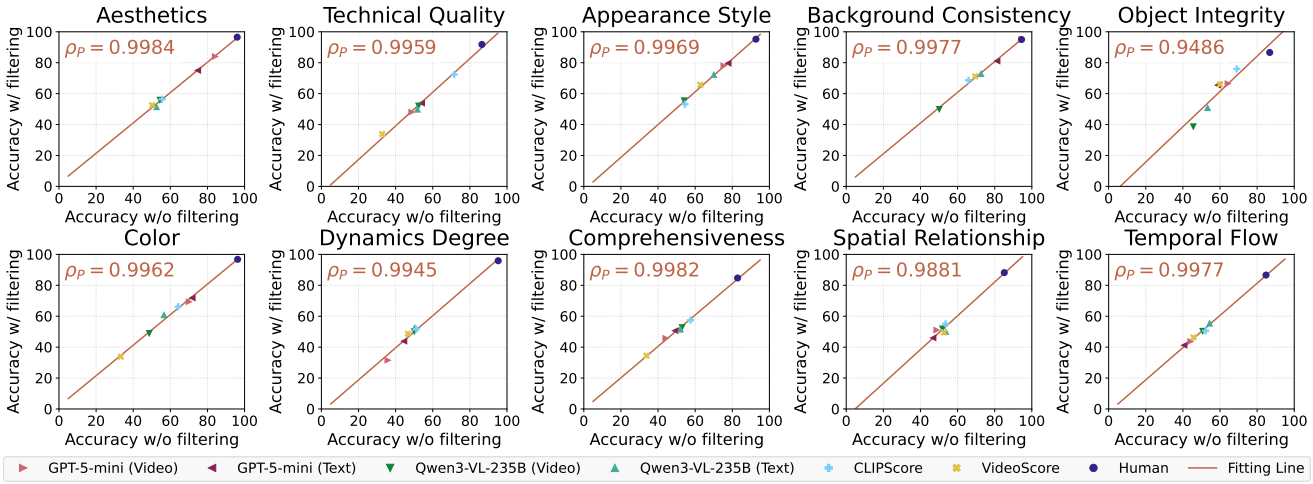

Figure 4. Relationship between accuracies before and after filtering on the degraded SLVMEval data. For each aspect, we plot the accuracy of every evaluation system before vs. after filtering and compute the Pearson correlation coefficient ρ_{P} from these points. Each marker corresponds to one evaluation system. The horizontal axis shows accuracy on the unfiltered data, and the vertical axis shows accuracy on the filtered test set. The solid line indicates the fitted linear regression line visualizing the correlation.

rather than a strong effect.[9]

Necessity of data cleaning. Figure 4 shows the Pearson correlation coefficient ρ_{P} between the accuracies obtained when evaluating each model on SLVMEval with manual data filtering and without manual data filtering. For all aspects, the correlations are consistently high (above 0.94). These results suggest that our degradation procedure can produce reasonably reliable video pairs for evaluation even without costly manual filtering. In turn, this implies that manual filtering is not strictly necessary when constructing domain-specific evaluation benchmarks or further extending our benchmark, highlighting the scalability of our benchmark construction method built on the degrading operations we introduce.

7. Conclusion

We introduced SLVMEval, a meta-evaluation benchmark for assessing T2LV evaluation systems. We adopted a pairwise comparison framework for evaluating videos. To create this benchmark, we start with a dense video captioning dataset and synthetically generate high-quality and low-quality video pairs. During meta-evaluation, the target evaluation system judges, for each video pair and specified aspect, which video is of higher quality. As a baseline, we conducted a meta-evaluation using existing evaluation systems. The results show that these systems underperform human performance in 9 out of 10 aspects, despite the task being easy for humans. Moreover, our findings suggest that the performance of evaluation systems tends to decline as video length increases, pointing to the limitations of existing systems in assessing T2LV. Furthermore, a comparison of evaluation results on SLVMEval with and without manual filtering suggests that, for the degradation operations we introduce, such filtering is not strictly necessary. This, in turn, indicates that our approach can serve as a scalable method for expanding the benchmark. Finally, the focus of our benchmark is to measure whether a system meets the minimum capabilities required of a T2LV evaluation system. We hope that future research on T2LV evaluation systems will use achieving human-level performance on our benchmark as an initial target.

Limitations. Our benchmark uses a synthetic dataset, so its distribution differs from the outputs expected from future T2LV models and may not cover all failures those models could produce. Therefore, additional benchmarks are required to evaluate models in more complex and realistic settings. Such benchmarks will need to be developed in parallel with advances in T2LV generation models, as their design depends on analyzing model failure patterns. We therefore anticipate further progress in this research area.

In addition, the settings of the degrading operations, such as the magnitude of contrast change and the fraction of clips degraded, act as hyperparameters. Varying these hyperparameters will change the difficulty of the benchmark. The benchmark is currently easy for humans to judge, which suits our evaluation goal. However, creating variants with different difficulties would enable broader analysis.

[9] We show the correlation coefficients and p-values for all systems and aspects in the supplementary material.

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